Standard Operating Procedure for Huawei Access Node (BAS-ACC-07) Optical Transport Failure (ALARM_4470)

Node: BAS-ACC-07 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_4470

Description

The Optical Transport Failure (ALARM_4470) on Huawei Access Node (BAS-ACC-07) at Basavanagudi POP is a critical fault that necessitates prompt attention to minimize downtime and ensure high network availability. This failure domain occurs due to the degradation of SFP/XFP transceivers or fiber cuts, leading to a loss of signal integrity on the optical transport links. The Huawei Access Node (BAS-ACC-07) operates as an aggregation point, connecting multiple access nodes to the core network. The optical transport links are critical for forwarding data between these nodes. When an optical transport link fails, it triggers the ALARM_4470 warning, indicating a potential loss of connectivity. To mitigate this risk, it is essential to identify and resolve the root cause of the failure promptly. A comprehensive understanding of the Huawei CLI dialect and the underlying network architecture is crucial to resolving this issue.

Pre-requisite Checks

- Check: Verify that the Basavanagudi POP is accessible via remote hands, Status: Completed

- Check: Confirm the status of redundant links to ensure no single point of failure, Status: Completed

- Check: Inspect the SFP/XFP transceivers for signs of degradation or physical damage, Status: Completed

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber cut	Medium
Rx/Tx power loss due to misconfigured optical parameters	Low

Diagnostic Commands & Expected Outputs

```
Command: huadisplay fault, Expected Output: Fault information for the optical transport links, Description: This command displays the status of optical transport links, including the SFP/XFP transceivers and fiber connections. It helps isolate the root cause of the ALARM_4470 warning.

Command: huadisplay interface brief, Expected Output: Interface status for the optical transport links, Description: This command provides a brief overview of the interface status, including the link speed,
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duplex mode, and error statistics. It aids in identifying the specific
interface causing the failure.
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Resolution Steps

Step 1: Step: Verify SFP/XFP transceiver status, Command: huadisplay transceiver, Description: This command displays the status of the SFP/XFP transceivers, including temperature, voltage, and optical parameters. It helps confirm if the transceiver is malfunctioning or requires replacement.

Step 2: Step: Swap SFP/XFP transceiver, Command: huaset transceiver swap, Description: If the SFP/XFP transceiver is found to be malfunctioning, this command swaps the transceiver with a spare one. It ensures minimal downtime and maintains network availability.

Post-Check Verification

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Command: huadisplay fault, Expected Output: Fault information for the
optical transport links (resolved), Description: This command verifies
that the ALARM_4470 warning has been cleared, indicating the
resolution of the optical transport failure.

Command: huadisplay interface brief, Expected Output: Interface status
for the optical transport links (resolved), Description: This command
confirms that the interface status has returned to normal, indicating
that the failure has been resolved.
```

Roll-back Procedures

- Step: Restore original SFP/XFP transceiver, Command: huaset transceiver restore, Description: If the fix fails, this command restores the original SFP/XFP transceiver, ensuring network stability and availability.